Project #2

1 Scope of the work

This project is an individual assignment aimed to study a material, structure, or system that is subjected to coupled boundary conditions (e.g., hydro-mechanical, thermo-hydraulic or thermo-mechanical conditions). The work presented shall contain a literature review, a simulation plan to test various scientific or technological hypotheses, Finite Element results to validate or invalidate these hypotheses, interpretations of potential numerical errors, and recommendations for future research or design. Students will propose a project topic and, after the instructor's approval, will use the computational software of their choice (e.g. MOOSE, MATLAB, ABAQUS, ANSYS, COMSOL, PLAXIS, OpenSees) to do the project. The project will be assessed based on a report, according to the rubrics explained in Section 2.

Requirements to comply with the course objectives are the following:

- The problem under study has to couple at least two degrees of freedom, e.g. hydro-mechanical, thermo-mechanical, hydro-chemical couplings.
- The study must be motivated by a research or design question, e.g. simulating a laboratory experiment to calibrate or validate a model, predicting the behavior of an engineered material.
- A sound literature review should justify the modeling approach proposed in the project (as a rule of thumb, 10 to 20 references are expected).
- It is necessary to design a research plan to test the hypotheses formulated, e.g. present the different models that will be used to understand processes that occur in an experiment, describe a simulation plan to compare different design approaches, explain how an existing numerical model will be enhanced to capture a given phenomenon, or justify a numerical benchmark that compares models, meshes and/or boundary conditions.
- All the balance and constitutive equations should be explained in detail in the report.
- Results should be interpreted to validate or invalidate the hypotheses formulated.
- A discussion of the results should examine the possible sources of approximation and errors in the simulations, and present some conclusions regarding the validity of the hypotheses formulated (e.g., explain which variables control the saturation propagation front around a nuclear waste disposal).

Example projects:

- A Finite Element approach for miscible contaminant fluid flow through deforming porous media.
- Numerical assessment of bioremediation to optimize soil tensile strength.
- A hydro-mechanical model for wicking in partially saturated porous media: towards applications of sorption predictions in concretes and mortars.
- Numerical hydro-mechanical modeling of the excavation of the tunnel for Changsha Metro Line 2.
- Numerical modeling of anisotropic propagation of hydraulic fractures in permeable and impermeable rock.
- Modeling of inter- and intra- granular crack propagation in salt rock during thermal consolidation.
- Non linear topology optimization of porous media under thermo-hydro-mechanical constraints.

2 Grading rubrics for the project report

Please prepare a report of no more than 20 pages including text, equations, tables, and figures, excluding the cover page and the references, using 1 inch margins and a 11pt font.

1. (10 pts) **Problem statement and proposed approach**. First, state the project motivation, scientific questions, unknown processes, computational challenges, or design issues. Then explain the objectives of the study and describe the computational tools employed (e.g., MOOSE? MATLAB? ANSYS?)

2. (10 pts) **Literature review** (in textbooks, journal articles, conference proceedings, technical reports and official websites). Explain the benefits and limitation of previous models. It is expected that you will cite at least 10-20 references outside of the course material. To cite a reference: search the reference in Google Scholar, click "cite" and copy-paste the full reference in the format of your choice (you can also use Endnote with Word, or Bibtex in LateX).

3. (20 pts) **Theoretical modeling**. Provide a description of the material of interest (at the scale of the Representative Elementary Volume), the momentum balance equations, the mass conservation equations, and any necessary state equations. Derive the governing equations to be used in the FEM (N independent PDEs to be solved for N field variables).

4. (20 pts) **Finite Element Method**. Present the variational formulation, the space and time discretization schemes and the elementary equations. You can use the matrix form of the equations to avoid lengthy derivations. Explain any algorithms/programs written for the project.

5. (20 pts) **Simulation results and interpretations**. Present the geometry, boundary conditions and loading path (stress, pore pressure, temperature, chemical concentration imposed). Then describe and interpret your results (from a physical standpoint). Answer the research/design questions raised in the problem statement, and provide the visualization necessary to support your interpretations.

6. (10 pts) **Numerical analysis**. Explain any source of error/approximation in your model. Assess the performance of you model if possible (e.g., through a calibration against a data set followed by a validation against an independent data set).

7. (10 pts) **Conclusions**. Based on the simulation results, recommend a design/modeling option in response to the problem, recommend further experimental tests or numerical analyses, and highlight the intellectual merit and broader impacts of the project.